

Announcing the Agentic Resource Discovery specification

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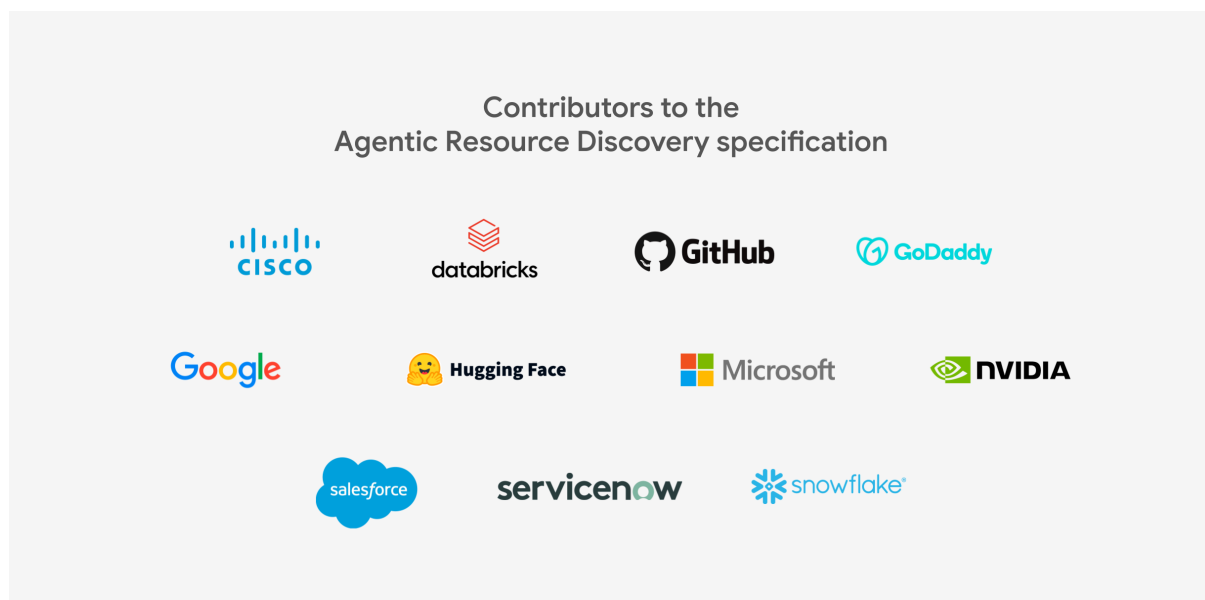
An open specification for finding and verifying tools, skills, and agents across the web.

Agents are becoming participants in a much larger ecosystem. They increasingly rely on tools, skills, and other agents distributed across teams, organizations, and

Today, there is no standard way to answer those questions across organizations.

That's why we're announcing **Agentic Resource Discovery (ARD)**, an open specification for publishing, discovering, and verifying AI capabilities across the web. Developed with partners across the industry, ARD allows tools and services to be securely shared and connected, regardless of their underlying framework, protocol, or provider.

Built with partners across the agent ecosystem



The missing layer of the agentic web

Consider an operations agent investigating a live production incident. To resolve the issue, it might need to query observability systems, search engineering documentation, review deployment history, open support tickets, and maybe even consult specialized troubleshooting agents.

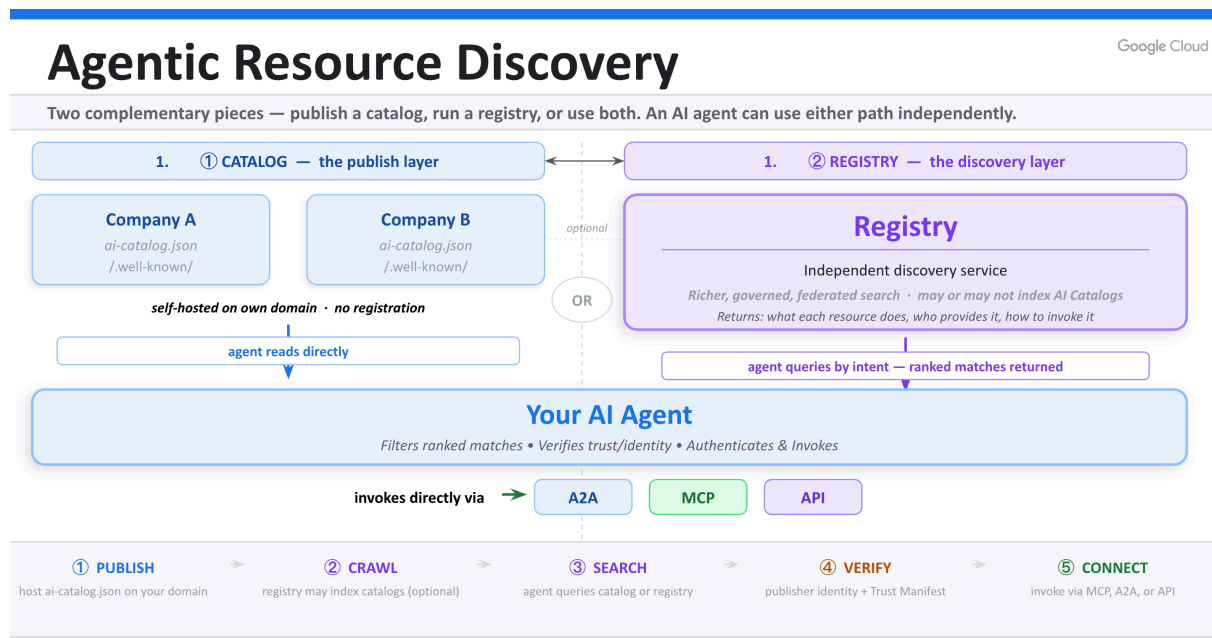
While many platforms already feature custom registries to manage these capabilities, they remain fragmented and siloed within specific ecosystems. This lack of interoperability prevents agents from easily communicating across

enabling any agent to dynamically find the right resources for the job. From there, ARD steps out of the way – handing off the verifiable trust metadata so the agent can establish a direct, secure connection using the tool's native protocol.

How ARD works

This architecture relies on two primitives: **catalogs** and **registries**.

- **Catalogs:** To make resources discoverable, an organization publishes a catalog describing its available capabilities. Because these catalogs are hosted directly under the organization's own domain, ownership of that domain serves as the cryptographic foundation for identity and trust.
- **Registries:** Registries act as search engines for the agentic web. They crawl published catalogs, index their contents, and make them searchable. When an agent submits a discovery request, a registry returns matching capabilities along with the metadata required to verify the publisher and establish trust before connecting.



The video below demonstrates an ARD client discovering and executing new capabilities entirely at runtime. Watch for these four key phases in the terminal

tools, or even other nested catalogs.

2. **Discovery and resolution:** When a client agent needs a capability, it can either query an ARD registry using a plain-language intent (which can actively crawl and index these catalogs), or it can completely bypass search and directly fetch a catalog from a known partner's domain.
3. **Cryptographic verification:** For production environments, the discovery layer allows publishers to attach verifiable trust metadata. Whether found via search or direct fetch, this enables the client agent or registry to actively confirm the publisher's true cryptographic identity before connecting to the endpoint.
4. **Direct runtime connection:** The client agent dynamically loads the selected capability, interacts with it using its native protocol or API, and returns the result to the user.

```
kchota-mac:ard-client kchota$ uv run --package travel-planner client
/Users/kchota/dev/ard-client/.venv/lib/python3.10/site-packages/google/ard/features/_feature_decorator.py:72: UserWarning: [EXPERIMENTAL] feature PLUGGABLE_AUTH is enabled.
check_feature_enabled()
ARD + ADK Runtime
ARD GENERIC RUNTIME CLIENT
+ Bootstrapping ARD Registry Client from http://127.0.0.1:8500/.well-known/ai-catalog.json...
+ Ready! Enter queries to search, resolve, and connect to agents.
Enter search query/task (or press Enter to exit client):
```

Bringing ARD to Gemini Enterprise Agent Platform



Agent Registry ensures enterprises can trust, govern, and operationalize that promise at scale. It provides fully hosted support for searching, discovering, and hosting agentic resources, including agents, skills, MCP servers and other tools. Additionally, it allows users to onboard these capabilities directly onto Agent Registry, and it will soon support authenticated publisher onboarding.

Importantly, Agent Registry plays a central role in enterprise governance. This includes assigning globally unique namespaced URNs, enforcing agentic egress policies, and pinning tools and specifications. It also handles secure resource management using [Agent Identity](#) to verify the [trust manifest](#) – ARD’s cryptographic layer for proving agent authenticity and meeting enterprise compliance standards like HIPAA.

Native support for ARD will be available in Agent Platform in the coming months, allowing organizations to securely connect their internal registries to the broader network.

Get started with ARD

The ARD specification is available now. To get involved:

- **Publish your first catalog:** Follow the [quickstart guide](#) to host an `ai-catalog.json` file on your domain and make your services discoverable in minutes.
- **Read the specification:** Check out the [specification](#), including full schemas, federation model, trust architecture, and reference implementations.
- **Join the community:** Contribute implementations, propose schema updates, and participate in the evolution of ARD by visiting our [GitHub repository](#).

The agent ecosystem works best when it is decentralized and open. ARD is our contribution to keeping it that way. We’d love your feedback – and your catalog!

Acknowledgements

The Agentic Resource Discovery specification is licensed under Apache 2.0 and is built upon the foundational AI Catalog data model. We are grateful to the [AI](#)

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MCP over HTTP
Host: mcp-server.example
X-MCP-Param-Project: myproject-prod
X-MCP-Param-Region: us-west1
X-MCP-Param-Instance: finance-db-01

Global Endpoint

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 "method": "spanner.execute_sql",
 "arguments": {
 "project": "myproject-prod",
 "region": "us-west1",
 "instance": "finance-db-01"
 }
}

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